

An Obstacle Avoiding Motion Planning Method for Manipulator using Progressive Dynamic Local Search

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Abstract—To enhance the efficiency of motion planning, we introduce a new angle profile searching method known as Progressive Dynamic Local Search (PDLS). Building upon a collision-free multi-segment path established by the RSPM-PS algorithm, PDLS methodically computes angle profiles from the initial to the target pose, focusing searches near each point's predecessor. It refines the path's smoothness by adding intermediate points and sets the local search radius through the Jacobian matrix, optimizing convergence. Our method also innovates in assessing angle profile quality, introducing a combined error function that judiciously weighs the importance of position and attitude accuracy. Furthermore, we mitigate sudden changes in joint acceleration by interpolating angle profiles using a quintic polynomial. Tests in MATLAB and on a JAKA Zu311 manipulator demonstrate PDLS's advantages: reduced run times, precise calculations, and improved convergence rates.

Keywords—angle profile planning, linear interpolation, robotic manipulator, obstacle avoiding motion planning, progressive dynamic local search (PDLS)

I. INTRODUCTION

Robotic manipulators, crucial in medical surgery [1], industry [2], agriculture [3], logistics [4], and marine exploration [5], comprise arms with rotational joints that manipulate tools to perform complex tasks. These joints' angles, known as angle profiles, define the tool's pose—its position and orientation—essential for precise motion. For brevity, we use **pose** to refer to the combination of **position and attitude**. In scenarios with obstacles, devising a motion plan that enables the end-effector to move from a start pose to a target pose without collision is a core challenge, especially since such plans must consider both the position and orientation of the end-effector to ensure accurate and safe maneuvering within the operational environment.

Numerous studies have targeted obstacle avoidance in robotic motion planning. Kang et al. [6] introduced a target-oriented sampling method, while Yuan et al. [7] and Zhang

et al. [8] improved rapidly-exploring random trees (RRT) for better convergence and efficiency. Gai et al. [9] and Chen et al. [10] utilized artificial potential fields (APF) for dynamic path planning, and Sepehri et al. [11] combined RRT and APF for comprehensive path and angle profile determination. Yang et al. [12] designed motion-specific reward functions for manipulator training via reinforcement learning. Li et al. [13] addressed the issues of recurrent neural network-based planning with a new recursive approach for more complete solutions.

Additionally, heuristic intelligent methods have also advanced obstacle avoidance in robotic motion planning [14–17]. Techniques include Cheng et al.'s [14] beetle antennae inspired search algorithm, Vazquez-Santiago et al.'s [15] genetic algorithms with enhanced sampling and kinematics, Cao et al.'s [16] multi-objective optimization for faster convergence, and Sahu et al.'s [17] refined firefly algorithm with a new error function for precise evaluations.

Despite the progress, issues like lengthy computation times and convergence challenges remain. To address these, we propose a novel method, Progressive Dynamic Local Search (PDLS). PDLS introduces a methodical approach to angle profile determination, focusing on search area optimization around preceding points, enhancing local search convergence with judicious interpolation and Jacobian-based adjustments, employing a dynamic error function for pose precision, and utilizing quintic polynomial interpolation to ensure smooth joint transitions.

II. PRELIMINARY MODEL OF FORWARD KINEMATICS OF THE MANIPULATOR

For a manipulator with n rotation joints, let θ as shown in (1) denotes an angle profile. We assume the joints are indexed from 1 at the base joint up to the end-effector joint.

$$\theta = [\theta_1, \theta_2, \dots, \theta_n] \quad (1)$$

With an angle profile θ and manipulator's standard Denavit-Hartenberg (DH) parameters, we can derive the transformation matrix $T_i^{(i-1)}$ from the $(i-1)$ -th joint's

This document is funded by Fujian Province University Industry-University Cooperation Project (2021H6030) and National Natural Science Foundation of China (62372190, 61972166).

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coordinate system to the i -th joint's, as (2) illustrates.

$$T_i^{(i-1)} = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \beta_i & \sin \theta_i \sin \beta_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \beta_i & -\cos \theta_i \sin \beta_i & a_i \sin \theta_i \\ 0 & \sin \beta_i & \cos \beta_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

where β_i , d_i , a_i are link angle, link offset and link length of joint i , respectively, $i \in \{1, 2, \dots, n\}$.

The end-effector's pose is given by the 4x4 matrix T_n^0 , comprising a position vector P and an attitude matrix R , as shown in (3).

$$T_n^0 = \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & P_x \\ R_{21} & R_{22} & R_{23} & P_y \\ R_{31} & R_{32} & R_{33} & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

The end-effector's pose matrix T_n^0 is determined by the angle profile θ and calculated using forward kinematics as (4) depicts. Similarly, joint i 's pose matrix T_i^0 is given by (5).

$$T_n^0 = f(\theta) = T_1^0 \times T_2^0 \times \dots \times T_n^{(n-1)} \quad (4)$$

$$T_i^0 = f(\theta) = T_1^0 \times T_2^0 \times \dots \times T_i^{(i-1)} \quad (5)$$

III. THE PDLS METHOD

PDLS method executes efficient motion planning in four key phases: beginning with 'End-effector Path Planning' to outline an initial trajectory, followed by 'Linear Interpolation' to adjust the trajectory for consistency, 'Angle Profile Planning' to determine precise joint orientations, and 'Angle Trajectory Smoothing' for fluid and continuous joint motion.

A. End-effector Path Planning

RSPM-PS algorithm [18] devises a collision-free path by charting a direct route and adding segmentation points to navigate around obstacles. It iterates to clarify the path and prunes excess points for succinctness, yielding an optimized trajectory as depicted in Fig. 1.

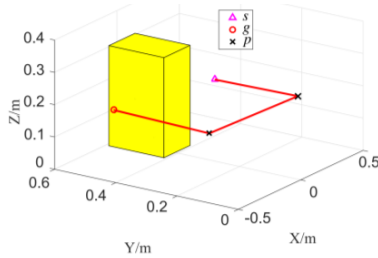


Fig. 1. A collision-free multi-segment path generated by RSPM-PS. s represents the initial point, g represents the target point, and p represents segmentation point.

B. Linear Interpolation

This step involves adding points to the path to maintain distances below the SPD threshold. The resulting multi-segment path, shown in Fig. 2. For specific end-effector attitudes at the target, we interpolate attitudes at each point, associating both position and attitude with a 4*4 matrix for each point in the path.

C. Angle Profile Planning

With $N+1$ points along the path, indexed from 0 (initial) to N (target), angle profiles are determined in sequence.

After establishing the initial point's profile, subsequent points are addressed sequentially, seeking each new angle profile close to its predecessor's. Local search size, θ_{size} , is crucial: too small risks failure, and too large wastes time.

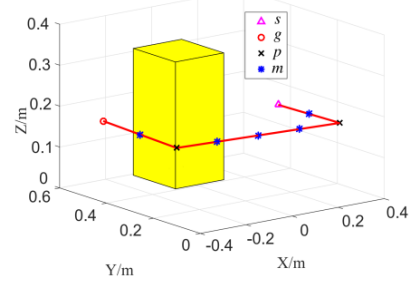


Fig. 2. A multi-segment path after linear interpolation. m represents interpolation points.

Determine local search size. The search size is calibrated using the Jacobian matrix pseudo-inverse to maintain a tight search range. If a search fails, which simulations show is uncommon, we increase the size dynamically with a scaling factor. This adaptive strategy introduces flexibility into the PDLS technique. Minor deviations in the angle profile θ result in proportional pose adjustments Δr , modeled linearly by the Jacobian matrix $J_r(\theta)$, as shown in (6).

$$\Delta r = J_r(\theta) \times \Delta \theta \quad (6)$$

Given the pose difference Δr , we set θ_{size} using (7), considering the preceding angle profile to ensure continuity across the trajectory.

$$\theta_{size} = |\Delta \theta| = |(J_r(\theta)^T J_r(\theta))^{-1} J_r(\theta)^T \times \Delta r| \quad (7)$$

Quality measure of angle profiles. Angle profile quality in local search is gauged by measuring the deviation of the end-effector's position and orientation from the target, using the 1-norm for position vectors and the Frobenius norm for orientation matrices, as outlined in (8) and (9).

$$\xi_{pos}(\theta_c) = \|f_{pos}(\theta_c) - T_{pos}\|_1 \quad (8)$$

$$\xi_{atti}(\theta_c) = \|f_{atti}(\theta_c) - T_{atti}\|_F \quad (9)$$

where $f_{pos}(\theta_c)$ and $f_{atti}(\theta_c)$ respectively represent the end-effector's position vector and orientation matrix determined by θ_c , and T_{pos} and T_{atti} represent the position and orientation(attitude) matrix of the current point.

The overall error function $\xi_{all}(\theta_c)$ combines these measures, adjusted by $\mu(i)$ and a weighting factor K , as outlined in (10) and (11). As the end-effector approaches the target, the importance of attitude gradually increases.

$$\xi_{all}(\theta_c) = \xi_{pos}(\theta_c) + \mu(i) * K * \xi_{atti}(\theta_c) \quad (10)$$

$$\mu(i) = \frac{1}{N+1-i} \quad (11)$$

Local search. In local search, we first determine the incremental change θ_{size} , sample and assess angle profiles

within the defined range, update the optimal profile θ_{best} , and use it to iteratively derive new profile θ_{new} by (12) to (14).

$$\theta_{rand}(i) = (2 \times rand(1) - 1) \times \theta_{size}(i) \quad (12)$$

$$\theta_{rand}(i) = \begin{cases} \theta_{rand}(i) \times \frac{\xi_{all}(\theta_{best})}{r_{th}}, & \xi_{all}(\theta_{best}) < \xi_{th, fine} \\ \theta_{rand}(i), & \xi_{all}(\theta_{best}) \geq \xi_{th, fine} \end{cases} \quad (13)$$

$$\theta_{new}(i) = \begin{cases} \theta_{max}(i), & \theta_{min}(i) + \theta_{rand}(i) > \theta_{max}(i) \\ \theta_{min}(i), & \theta_{min}(i) + \theta_{rand}(i) < \theta_{min}(i) \\ \theta_{min}(i) + \theta_{rand}(i), & \text{else} \end{cases} \quad (14)$$

In (12), random function $rand(1)$ selects a value uniformly in range $[0,1]$. In (13), $\xi_{th, fine}$ is a threshold used to fine the changes in angle profile when the end-effector's pose of the best angle profile is very close to the current point. In (14), $\theta_{min}(i)$ and $\theta_{max}(i)$ represent the lower limit and upper limit of the joint angle of joint i , $i \in \{1, 2, \dots, n\}$.

For θ_{new} , collision checks are essential. Collision detection is streamlined by representing manipulator links as cylinders and obstacles as cubes, as shown in Fig.3. The shortest distance d from the center line of link to an obstacle is computed through vertical projection from the center line of the link to each surface of the obstacle. If d below a critical value, indicates a collision.

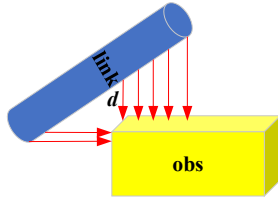


Fig. 3. Collision detection of a link of the manipulator with an obstacle.

The search iterates until the overall error function $\xi_{all}(\theta_{best})$ meets a defined convergence precision $\xi_{th, conv}$, or until a set number of iterations $ITER$ is reached. If the error remains above $\xi_{th, conv}$, we enlarge the search size θ_{size} to $\kappa\theta_{size}$ to improve chances of finding a valid profile.

D. Angle Trajectory Smoothing

Joint motions are smoothed using a quintic polynomial interpolation to prevent sudden changes in acceleration, expressed in (15).

$$\theta_i(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \quad (15)$$

where $a_0, a_1, a_2, a_3, a_4, a_5$ are parameters to be determined. For two adjacent points, let assume the time corresponding to the first point is t_0 , and the time corresponding to the second point is t_f . To ensure smooth interpolation, we impose 6 constraints on joint angle, joint rotation velocity and joint rotation acceleration, as shown in (16).

$$\begin{cases} \theta_i^0 = a_0 \\ \dot{\theta}_i^0 = a_0 + a_1 t_f + a_2 t_f^2 + a_3 t_f^3 + a_4 t_f^4 + a_5 t_f^5 \\ \ddot{\theta}_i^0 = a_1 \\ \dot{\theta}_i^f = a_1 + 2a_2 t_f + 3a_3 t_f^2 + 4a_4 t_f^3 + 5a_5 t_f^4 \\ \ddot{\theta}_i^f = 2a_2 \\ \ddot{\theta}_i^0 = 2a_2 + 6a_3 t_f + 12a_4 t_f^2 + 20a_5 t_f^3 \end{cases} \quad (16)$$

where $\theta_i^0, \dot{\theta}_i^0$ and $\ddot{\theta}_i^0$ are joint angle, joint rotation velocity and joint rotation acceleration of joint i at time 0, respectively. $\theta_i^f, \dot{\theta}_i^f$ and $\ddot{\theta}_i^f$ are joint angle, joint rotation velocity and joint rotation acceleration of joint i at time t_f , respectively. The solution of (16) is shown in (17).

$$\begin{cases} a_0 = \theta_i^0 \\ a_1 = \dot{\theta}_i^0 \\ a_2 = \ddot{\theta}_i^0 / 2 \\ a_3 = 20\theta_i^f - 20\theta_i^0 - (80\dot{\theta}_i^f + 12\dot{\theta}_i^0)t_f - (3\ddot{\theta}_i^0 - \ddot{\theta}_i^f)t_f^2 / 2t_f^3 \\ a_4 = 30\theta_i^0 - 30\theta_i^f + (14\dot{\theta}_i^f + 16\dot{\theta}_i^0)t_f + (3\ddot{\theta}_i^0 - 2\ddot{\theta}_i^f)t_f^2 / 2t_f^4 \\ a_5 = 12\theta_i^f - 12\theta_i^0 - (60\dot{\theta}_i^f + 60\dot{\theta}_i^0)t_f - (\ddot{\theta}_i^0 - \ddot{\theta}_i^f)t_f^2 / 2t_f^5 \end{cases} \quad (17)$$

Given $\dot{\theta}_i^0, \ddot{\theta}_i^0, \dot{\theta}_i^f$ and $\ddot{\theta}_i^f$, parameters a_0 through a_5 are derived using (17). Here we set the values of $\dot{\theta}_i^0, \ddot{\theta}_i^0$ for all the points in the path using (18), where d_{i-1} represents the distance between the $(i-1)$ -th point and the i -th point.

$$\begin{cases} \dot{\theta}_i = \begin{cases} 0, & i = 1 \text{ or } N \\ (d_{i-1} + d_i) / 2t_f, & 1 < i < N \end{cases} \\ \ddot{\theta}_i = 0 \end{cases} \quad (18)$$

With above preparations, the parameters a_0 through a_5 can be obtained straightforwardly using (17). Then we can use (15) to determine the angles of some additional intermediate points.

IV. PERFORMANCE EVALUATION

A. Target Manipulator

Performance of the PDLs method was evaluated through simulations and real-world tests using a JAKA Zu311 manipulator, demonstrating enhanced efficiency and precision. The JAKA Zu311 manipulator is shown in Fig.4, and its standard DH parameters are shown in Table I.

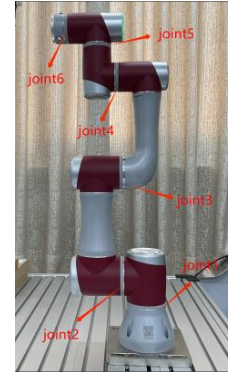


Fig. 4. JAKA Zu311 robotic manipulator.

TABLE I. STANDARD DH PARAMETERS OF THE JAKA ZU311 MANIPULATOR.

Joint	θ_i (radian)	d_i (m)	a_i (m)	β_i (radian)
1	θ_1	0.15055	0	1.5708
2	θ_2	0	0.246	0
3	θ_3	0	0.19004	0
4	θ_4	-0.11621	0	1.5708
5	θ_5	0.1175	0	-1.5708
6	θ_6	0.105	0	0

B. Simulation Experiments

We compare PDLS performance against APF [13], BAS [14] and RRT [8], including variants without the linear interpolation step (NoLI). Tests are run in an environment with an obstructing cube, as shown in Fig. 5, where the yellow cube represents an obstacle, the initial point and the target point are located on both sides of the obstacle.

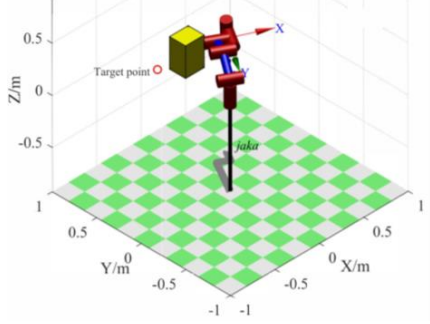


Fig. 5. Experiment environment and the initial pose of the manipulator.

The angle profile θ_0 for the initial point is: $[-125^\circ, 161^\circ, 21.3^\circ, 72.5^\circ, -93.9^\circ, -0.006^\circ]$, and the target point has position and attitude is expressed as matrix T_{arg} :

$$T_{\text{arg}} = \begin{bmatrix} 0.5612 & 0.8075 & -0.1817 & -0.4038 \\ 0.8251 & -0.5283 & 0.2002 & 0.4117 \\ 0.0657 & -0.2622 & -0.9628 & 0.1527 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

The experiment results of the eight methods are shown in Table II. PDLS outperforms in speed and trajectory optimization, highlighting the linear interpolation step's importance, which is essential for algorithm convergence and smooth path generation.

TABLE II. RESULTS OF COMPARATIVE EXPERIMENTS.

Method	Running time (s)	Length of trajectory (m)
PDLS	33.7595	1.0955
PDLS-NoLI	INF	INF
APE	60.3426	1.9541
APE-NoLI	INF	INF
BAS	181.9042	2.0354
BAS-NoLI	INF	INF
RRT	770.0132	8.7372
RRT-NoLI	INF	INF

C. Motion Planning

In the Fig.5 setup, PDLS facilitates the manipulator's motion planning. In this experiment, SPD threshold is 0.20m, r_{th} is 0.05, R_{th} is 0.01, the convergence precision is 0.1, the maximum number of searches $ITER$ is 50, and the scaling factor κ is 1.2.

The effectiveness of PDLS is evident in Table III, where all ξ_{all} values fall below the convergence precision, indicating reliable algorithm performance. The PDLS method is both precise and swiftly convergent, as the running time marginally increases to 43.1672 seconds when setting R_{th} to 0.005.

TABLE III. ξ_{all} VALUES OF ANGLE PROFILES FOR ALL POINTS IN THE PATH.

index of point	0	1	2	3	4	5	6	7	8
ξ_{all}	0	0.0092	0.0056	0.0077	0.0079	0.0071	0.0082	0.0097	0.0072

Collision detection is crucial in motion planning to ensure the manipulator operates safely. Table IV details the minimum distances between the manipulator's links and any obstacles during its movement, with a safety margin set at greater than 0.1 meters. The data confirms that the manipulator's links maintain this safe distance, avoiding collisions throughout the operation.

TABLE IV. MINIMUM DISTANCE BETWEEN JOINTS AND OBSTACLE DURING MOVEMENT.

index of journey	1	2	3	4	5	6	7	8
Link1	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
Link2	0.183	0.262	0.311	0.365	0.348	0.316	0.230	0.218
Link3	0.106	0.455	0.102	0.128	0.197	0.280	0.309	0.300
Link4	0.306	0.360	0.140	0.205	0.140	0.296	0.234	0.212
Link5	0.369	0.394	0.316	0.112	0.103	0.235	0.198	0.108
Link6	0.298	0.318	0.266	0.128	0.101	0.235	0.198	0.183

The quintic polynomial ensures a smooth angular trajectory, preventing abrupt rotation accelerations and ensuring steady end-effector movement. Fig.6(a) clearly reveals that there are sudden changes in joint rotation acceleration at the points, while Fig.6(b) displays the position changes of the end-effector during movement. By enabling angle trajectory smoothing, we obtain improved results as shown in Fig.6(c) and Fig.6(d). Obviously, sudden changes in rotation acceleration of the joints are avoided.

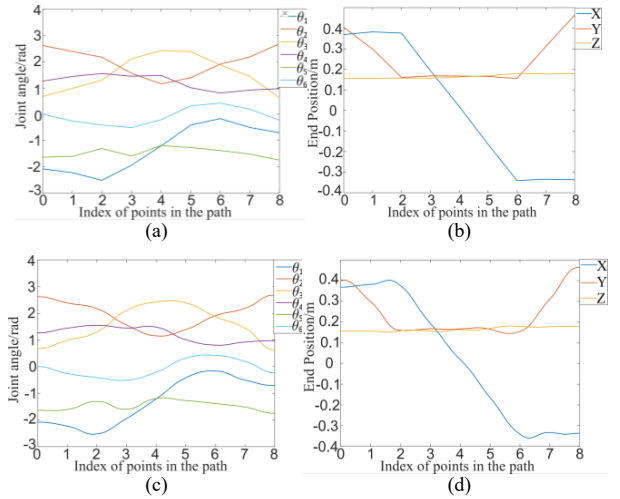


Fig. 6. Comparison of smoothed results for angle trajectory.

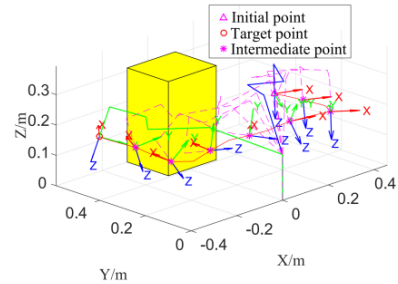


Fig. 7. Poses of the manipulator for all points in the path.

Fig. 7 shows the manipulator's real movement path and poses at nine points along it. The initial pose is marked by a blue solid line, the target pose by a green solid line, and intermediate poses by pink dotted lines. The trajectory of the end-effector from start to end is represented by the red solid line. The XYZ axes are displayed on the end-effector's

position to indicate its orientation, where the Z-axis represents the direction of the end-effector.

D. Experiments on Real Manipulator

The PDLS was tested on JAKA Zu311 manipulator. Fig.8(a) provides the scene of our experiment, where there is a carton as an obstacle, the initial point and the target point are as indicated. Some snap shots of the system when the manipulator is moving are shown in Fig.8. Fig.8(b) and Fig.8(c) respectively shows the poses of the manipulator corresponding to the 2nd and 6th points of the path. Fig.8(d) demonstrates the target pose of the manipulator. The experiment shows that, following the angle sequences determined by the PDLS method, the manipulator drives its end-effector to move successfully and collision-free from the initial point to the target point with required attitude at the target point.

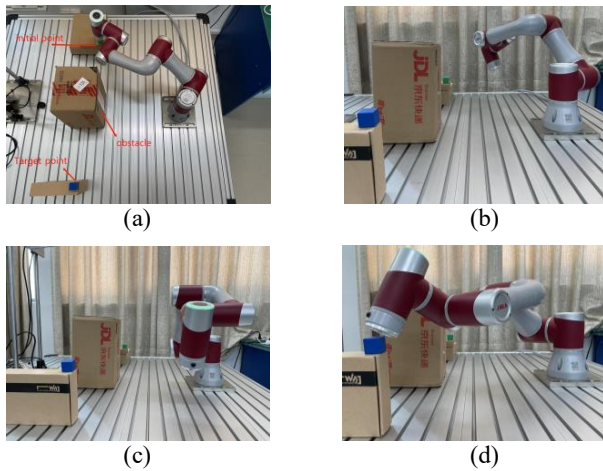


Fig. 8. Results of experiment on real manipulator.

V. CONCLUSION

This paper introduces a novel obstacle-avoiding motion planning method called Progressive Dynamic Local Search (PDLS). The PDLS method formulates a collision-free plan through four main steps. Initially, it generates a collision-free multi-segment path from the initial to the target position amidst obstacles. Subsequently, linear interpolation inserts additional points to maintain distances between points below the SPD threshold. PDLS then sequentially calculates angle profiles from the initial to the target pose, searching near the prior point's profile for the next one. To enhance local search convergence, PDLS limits the distance between path points through interpolation and adjusts the search radius using the Jacobian matrix. A unique combined error function evaluates the angle profiles by balancing the importance of position and attitude accuracy. Finally, to prevent sudden joint acceleration changes, it interpolates angle profiles using a quintic polynomial function. The method's efficacy, demonstrated in experiments, shows PDLS's advantage in creating smooth, precise, and collision-free motion plans compared to traditional methods, proving its effectiveness in environments with known obstacles.

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